COMPOSITIONS OF LOCAL COMPLETE INTERSECTION MORPHISMS OF SCHEMES ARE LOCAL COMPLETE INTERSECTION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 18207-18211 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 18213-18242, the complete author proof (28 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Counted A side only: the spaces twin's proof is incomplete per ruling v3. Pillar of L16201's precomposition stability.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Divisors, lemmas regular-immersion-noetherian and composition-regular-immersion; Morphisms, lemma composition-smooth; chapter definition-lci and its affine-space normalization discussion.

Source reference. Original lemma 069F: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18111–18121. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 067Q: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18242–18242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 063L: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18242–18242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01VA: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 18242–18242. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A composition of local complete intersection morphisms is a local complete intersection morphism.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. Let $g : Y \to S$ and $f : X \to Y$ be local complete intersection morphisms. Let $x \in X$ and set $y = f(x)$. Choose an open neighbourhood $V \subset Y$ of y and a factorization $g|_V = \pi \circ i$ for some Koszul-regular immersion $i : V \to P$ and smooth morphism $\pi : P \to S$. Next choose an open neighbourhood U of $x \in X$ and a factorization $f|_U = \pi' \circ i'$ for some Koszul-regular immersion $i' : U \to P'$ and smooth morphism $\pi' : P' \to Y$. In fact, we may assume that $P' = \mathbf{A}^n_V$, see discussion preceding and following Definition 069F. Picture:

$$\begin{array}{ccccc} X & \longleftarrow & U & \xrightarrow[i']{} & P' = \mathbf{A}^n_V \\ \downarrow & & & & \downarrow \\ Y & \longleftarrow & & & V \xrightarrow[i]{} P \\ \downarrow & & & & \downarrow \\ S & \longleftarrow & & & S \end{array}$$

Set $P'' = \mathbf{A}^n_P$. Then $U \to P' \to P''$ is a Koszul-regular immersion as a composition of Koszul-regular immersions, namely i' and the flat base change of i via $P'' \to P$, see Divisors, Lemma 063L and Divisors, Lemma 067Q. Also $P'' \to P \to S$ is smooth as a composition of smooth morphisms, see Morphisms, Lemma 01VA. Hence we conclude that $X \to S$ is Koszul at x as desired. □